

MTT-I (SET-B)
Academic Year 2024-25 (Odd Semester)

Course	:	B.Tech.	Date	:	7/11/2024
Semester/ Section	:	First/L-V	Time Duration	:	1:30 Hrs.
Subject & Subject Code	:	Engineering Chemistry (1FY2-03)	Max. Marks	:	50

Course Outcomes

CO1	Explain the impurities of water (mainly hardness) and boiler troubles.
CO2	Describe processing technologies of fuel with numerical aspects of combustion of fuel.

Instructions: Attempt all sections.

Q. No.	CO	Questions	Marks
PART-A Attempt All Questions (10x2 = 20 Marks)			
1.	1	Define the term hardness with the help of chemical reaction.	2
2.	1	Explain caustic embrittlement in boilers?	2
3.	1	Why do we express hardness in terms of CaCO_3 equivalent?	2
4.	1	Define sterilization?	2
5.	1	Explain Break point chlorination with the help of graph.	2
6.	2	Define LCV. and HCV	2
7.	2	Enlist any five characteristics of an ideal fuel?	2
8.	2	Define the term cracking?	2
9.	2	Define the term octane number.	2
10.	2	Name any two anti-knocking agents.	2

PART-B

(Descriptive /Analytical / Problem Solving/ Design questions)

Attempt Any Three Questions (3x5=15 Marks)

1.	1	Explain softening of water? Describe softening of water by ion exchange method with diagram.	5
2.	1	Define the term boiler trouble and explain scale formation and its prevention (any two methods) with chemical reactions.	5
3.	1	Standard hard water contains 1g of CaCO_3 per litre. 100 ml of standard hard water requires 25ml of EDTA solution. 100ml of sample water required 18 ml of EDTA solution. The sample water after boiling required 12ml of EDTA solution. Calculate temporary, permanent and total hardness of sample water	5
4.	1	Calculate the quantity of hydrated lime and sodium carbonate required for softening 10,000 litres of water containing the following impurities: $\text{CaCO}_3 = 10.0 \text{ mg/L}$; $\text{Mg}(\text{HCO}_3)_2 = 14.6 \text{ mg/L}$; $\text{CaCl}_2 = 11.1 \text{ mg/L}$; $\text{MgSO}_4 = 6.0 \text{ mg/L}$; $\text{NaCl} = 1.0 \text{ mg/L}$. The purity of lime is 90 percent and that of sodium carbonate is 85 percent.	5

PART-C

(Descriptive /Analytical / Problem Solving/ Design questions)

Attempt Any Three Questions (3x5=15 Marks)

1.	2	Define HCV & LCV. Explain how Calorific value of solid and liquid fuel is determined by Bomb Calorimeter.	5
2.	2	A sample of coal was found to contain C=80%, H=7%, O=3%, S=2% and rest is ash. Calculate HCV & LCV of this coal sample using Dulong's formula.	5
3.	2	Explain Fischer-Tropsch Method of preparing synthetic petrol.	5
4.	2	The percentage composition of a coal sample is C = 80%, H = 4%, O = 3%, N = 3%, S = 2% and ash = 4.5%. Calculate the quantity of air required by weight, for the complete combustion of 1kg of 60% excess air is supplied	5